CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

The reflexive completion in context

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Abstract

For a small category A, Isbell conjugacy defines an adjunction between the functor categories [A^op, Set] and [A, Set]^op. The invariant part of this adjunction (that is, its category of fixed points) is called the reflexive completion of A. I will describe some results relating the reflexive completion to other kinds of completion. For example, there is a precise sense in which it is the intersection of the two functor categories above. The reflexive completion is unlike many other completions, in that it is not the completion with respect to any class of limits or colimits. It is functorial, but only in a rather subtle and unusual sense, as I will explain.